SUPPLY CHAIN ANALYTICS

Dataset Overview • SQL Analysis & Business Insights • Final Recommendations

Source: DataCo Global Supply Chain Dataset and the supplied project report. Findings and SQL are organized from the source document; validate calculated results against the original dataset before external publication.

PART 1 — DATASET OVERVIEW

1.1 Project and business context

The project analyzes DataCo Global's end-to-end order fulfilment operations, from order placement through shipping and delivery. The analysis focuses on delivery performance, profitability, customer and market performance, discounting, fraud-risk signals, data quality, and management KPIs.

1.2 Dataset profile

Attribute	Details
Dataset	DataCo Global Supply Chain Dataset
Records	180,519 rows
Original columns	53; 47 retained after documented cleaning
Unique orders	65,752
Unique customers	20,652
Time period	January 2015 – February 2018
Markets	LATAM, Europe, Pacific Asia, USCA, Africa
Products	118 unique products / SKUs across 50 categories
Main analytical tools	Python, Pandas, SQL, exploratory data analysis and BI reporting

1.3 Data domains and analytical use

Data domain	Representative fields / purpose
Order and dates	Order ID, order date, shipping date; time trends and order-level analysis
Customer	Customer ID, customer segment, city/state/country; segmentation and geographic analysis
Product and category	Product name, category, department, quantity and price; SKU/category performance
Financial	Sales, order item total, profit per order, discount rate, profit ratio; revenue and margin analysis
Logistics	Shipping mode, actual shipping days, scheduled shipping days, delivery status and late-delivery risk; SLA performance
Payment and order status	Payment type, order status; payment mix, cancellations, pending orders and fraud signals
Geography	Market, region, country and city; regional benchmarking

1.4 Data quality assessment

Issue in source report	Scale / finding	Treatment or implication
Product Description entirely null	180,519 rows	Drop column; no descriptive information available
Order Zipcode missing	86.2% missing (155,679 rows)	Drop as unreliable for granular geographic analysis
Customer Email placeholder	One unique value	Drop; not usable for customer outreach

Customer Password placeholder	One unique value	Drop due to security and lack of analytical value
Product Image	URL strings	Drop from analytical model
Product Status	Zero variance (all values 0)	Drop constant field
Customer last name missing	8 rows	Fill with “Unknown” to preserve rows
Customer zipcode missing	3 rows	Source report fills with 0; treat as a placeholder, not a valid postal code
Duplicate rows	0 reported	No duplicate-row removal required
Negative-profit orders	33,784 (18.7%)	Retain and flag for profitability analysis

1.5 Cleaning and feature engineering

- Removed low-value, constant, placeholder, security-sensitive, and heavily missing columns as documented.
- Converted order and shipping date fields to datetime for time-series analysis.
- Handled the small number of missing customer fields without dropping otherwise usable records.
- Standardized text categories and renamed fields for Python/SQL usability.
- Created analytical features including delivery_delay_days, is_late_delivery, profit_margin_pct, is_high_value, order year/month/quarter, discount_band, shipping_speed, risk_category, and revenue_per_unit.

1.6 KPI baseline in the source report

KPI	Reported value	Target / benchmark in source
Total revenue	$36.78M	—
Total profit	$3.97M	—
Profit margin	10.8%	>15%
Late delivery rate	57.3%	<20%
Average shipping days	3.5 days	3.0 days
Suspected fraud rate	2.25%	<0.5%
Negative-profit rate	18.7%	<5%
Unique customers	20,652	+15% YoY growth target

PART 2 — SQL QUERIES AND BUSINESS INSIGHTS

The source document includes three SQL levels: basic aggregations, intermediate business analysis, and advanced window-function/CTE analysis. Each query below is paired with the business insight recorded in the source report.

LEVEL 1 — Basic SQL

1. Query 1: What is the total revenue across all orders?

```
SELECT SUM(Sales) AS total_revenue
FROM supply_chain_orders;
```

Business insight: Baseline revenue: $36.78M total

2. Query 2: How many unique orders exist in the dataset?

```
SELECT COUNT(DISTINCT Order_Id) AS unique_orders
FROM supply_chain_orders;
```

Business insight: 65,752 unique orders across 4 years

3. Query 3: What are the distinct delivery statuses?

```
SELECT DISTINCT Delivery_Status
FROM supply_chain_orders
ORDER BY Delivery_Status;
```

Business insight: Reveals 4 categories: Late, On Time, Advance, Canceled

4. Query 4: Show top 10 orders by profit

```
SELECT Order_Id, Sales, Order_Profit_Per_Order
FROM supply_chain_orders
ORDER BY Order_Profit_Per_Order DESC
LIMIT 10;
```

Business insight: Identify highest value individual transactions

5. Query 5: How many orders were placed per market?

```
SELECT Market, COUNT(DISTINCT Order_Id) AS order_count
FROM supply_chain_orders
GROUP BY Market
ORDER BY order_count DESC;
```

Business insight: LATAM leads with most orders; Africa has fewest

6. Query 6: List all shipping modes and their frequency

```
SELECT Shipping_Mode, COUNT(*) AS order_count
FROM supply_chain_orders
GROUP BY Shipping_Mode
ORDER BY order_count DESC;
```

Business insight: Standard Class dominates at ~60% of orders

7. Query 7: Which customer segments exist?

```
SELECT Customer_Segment, COUNT(DISTINCT Customer_Id) AS customers
FROM supply_chain_orders
GROUP BY Customer_Segment;
```

Business insight: 3 segments: Consumer, Corporate, Home Office

8. Query 8: What is the average order value by payment type?

```
SELECT Type AS payment_type,
  ROUND(AVG(Sales), 2) AS avg_sales
FROM supply_chain_orders
GROUP BY Type
ORDER BY avg_sales DESC;
```

Business insight: All payment types show similar avg order values ~$204

9. Query 9: What percentage of orders used each payment type?

```
SELECT Type,
  COUNT(*) AS count,
  ROUND(COUNT(*)*100.0/(SELECT COUNT(*) FROM supply_chain_orders), 1) AS pct
FROM supply_chain_orders
GROUP BY Type;
```

Business insight: Debit is the most common payment at ~38%; Cash is least at ~11%

LEVEL 2 — Intermediate SQL

10. Query 1: What is the late delivery rate by shipping mode?

```
SELECT Shipping_Mode,
  COUNT(*) AS total_orders,
  SUM(CASE WHEN Delivery_Status = 'Late delivery' THEN 1 ELSE 0 END) AS late_orders,
  ROUND(SUM(CASE WHEN Delivery_Status = 'Late delivery' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 1) AS late_pct
FROM supply_chain_orders
GROUP BY Shipping_Mode
ORDER BY late_pct DESC;
```

Business insight: Same Day has the highest late delivery % despite being premium

11. Query 2: Monthly revenue trend

```
SELECT DATE_FORMAT(order_date, '%Y-%m') AS month,
  ROUND(SUM(Sales)/1000000, 2) AS revenue_millions
FROM supply_chain_orders
GROUP BY month
ORDER BY month;
```

Business insight: Reveals seasonality peaks in Q4 each year

12. Query 3: Profit by customer segment

```
SELECT Customer_Segment,
  ROUND(SUM(Order_Profit_Per_Order), 0) AS total_profit,
  ROUND(AVG(Order_Profit_Per_Order), 2) AS avg_profit
FROM supply_chain_orders
GROUP BY Customer_Segment
ORDER BY total_profit DESC;
```

Business insight: Consumer segment drives most total profit despite lower avg

13. Query 4: Orders with negative profit — fraud suspects

```
SELECT Order_Id, Sales, Order_Profit_Per_Order, Order_Status
FROM supply_chain_orders
WHERE Order_Profit_Per_Order < 0
  AND Order_Status = 'SUSPECTED_FRAUD'
ORDER BY Order_Profit_Per_Order ASC
LIMIT 20;
```

Business insight: Fraud orders often have the deepest negative profits

14. Query 5: Top 10 products by revenue

```
SELECT Product_Name,
  ROUND(SUM(Sales)/1000, 1) AS revenue_k,
  COUNT(DISTINCT Order_Id) AS order_count
FROM supply_chain_orders
GROUP BY Product_Name
ORDER BY revenue_k DESC
LIMIT 10;
```

Business insight: Field & Stream, Perfect Fitness lead revenue charts

15. Query 6: Discount rate vs profit — correlation check

```
SELECT
  CASE
    WHEN Order_Item_Discount_Rate = 0 THEN "No Discount"
    WHEN Order_Item_Discount_Rate <= 0.1 THEN "Low"
    WHEN Order_Item_Discount_Rate <= 0.2 THEN "Medium"
    ELSE "High"
  END AS discount_band,
  ROUND(AVG(Order_Profit_Per_Order), 2) AS avg_profit
FROM supply_chain_orders
GROUP BY discount_band
ORDER BY avg_profit DESC;
```

Business insight: Higher discounts sharply reduce or eliminate profit

16. Query 7: Sales and profit by order quarter

```
SELECT order_year,
  QUARTER(order_date) AS quarter,
  ROUND(SUM(Sales)/1000000, 2) AS sales_M,
  ROUND(SUM(Order_Profit_Per_Order)/1000, 1) AS profit_K
FROM supply_chain_orders
```

```
GROUP BY order_year, quarter
ORDER BY order_year, quarter;
```

Business insight: Q4 consistently outperforms other quarters

17. Query 8: Average shipping delay by region

```
SELECT Order_Region,
  ROUND(AVG(Days_for_shipping_real - Days_for_shipment_scheduled), 2) AS avg_delay_days
FROM supply_chain_orders
GROUP BY Order_Region
ORDER BY avg_delay_days DESC
LIMIT 10;
```

Business insight: Some regions face chronic shipping delays > 1 day avg

18. Query 9: Revenue breakdown by category

```
SELECT Category_Name,
  COUNT(DISTINCT Order_Id) AS orders,
  ROUND(SUM(Sales)/1000, 1) AS revenue_K
FROM supply_chain_orders
GROUP BY Category_Name
ORDER BY revenue_K DESC
LIMIT 15;
```

Business insight: Fishing, Cleats, Camping dominate category revenue

19. Query 10: Cancellation rate by market

```
SELECT Market,
  COUNT(*) AS total,
  SUM(CASE WHEN Order_Status IN ('CANCELED','ON_HOLD') THEN 1 ELSE 0 END) AS problem_orders,
  ROUND(SUM(CASE WHEN Order_Status IN ('CANCELED','ON_HOLD') THEN 1 ELSE 0 END)*100.0/COUNT(*), 1) AS
cancel_rate_pct
FROM supply_chain_orders
GROUP BY Market
ORDER BY cancel_rate_pct DESC;
```

Business insight: High cancel rates signal fulfillment problems or customer experience issues

20. Query 11: Department revenue and profit performance

```
SELECT Department_Name,
  COUNT(DISTINCT Order_Id) AS orders,
  ROUND(SUM(Sales)/1000000, 2) AS revenue_M,
  ROUND(SUM(Order_Profit_Per_Order)/1000, 1) AS profit_K,
  ROUND(SUM(Order_Profit_Per_Order)/SUM(Sales)*100, 1) AS margin_pct
FROM supply_chain_orders
GROUP BY Department_Name
ORDER BY revenue_M DESC;
```

Business insight: Fan Shop, Apparel lead department revenue

21. Query 12: Average items per order by segment

```
SELECT Customer_Segment,
  ROUND(AVG(Order_Item_Quantity), 2) AS avg_qty_per_item,
  ROUND(AVG(Order_Item_Total), 2) AS avg_item_value
FROM supply_chain_orders
GROUP BY Customer_Segment;
```

Business insight: Corporate buys larger quantities; Home Office has highest avg item value

22. Query 13: Shipping days — actual vs scheduled comparison

```
SELECT Shipping_Mode,
  ROUND(AVG(Days_for_shipment_scheduled), 1) AS scheduled_days,
```

```
  ROUND(AVG(Days_for_shipping_real), 1) AS actual_days,
  ROUND(AVG(Days_for_shipping_real - Days_for_shipment_scheduled), 2) AS avg_delay
FROM supply_chain_orders
GROUP BY Shipping_Mode
ORDER BY avg_delay DESC;
```

Business insight: All shipping modes run over schedule on average; Standard Class worst

23. Query 14: Which order statuses have the highest avg sales value?

```
SELECT Order_Status,
  COUNT(DISTINCT Order_Id) AS orders,
  ROUND(AVG(Sales), 2) AS avg_sales
FROM supply_chain_orders
GROUP BY Order_Status
ORDER BY avg_sales DESC;
```

Business insight: COMPLETE orders have highest avg sales; CANCELED are lowest — selection effect

LEVEL 3 — Advanced SQL (CTEs, Window Functions)

24. Query 1: Running total of monthly revenue (Window Function)

```
WITH monthly AS (
  SELECT DATE_FORMAT(order_date, "%Y-%m") AS month,
    SUM(Sales) AS monthly_revenue
  FROM supply_chain_orders
  GROUP BY month
)
SELECT month, monthly_revenue,
  SUM(monthly_revenue) OVER (ORDER BY month) AS running_total
FROM monthly
ORDER BY month;
```

Business insight: Tracks cumulative revenue growth over time — useful for annual target tracking

25. Query 2: Rank products by revenue within each category

```
WITH product_sales AS (
  SELECT Category_Name, Product_Name, SUM(Sales) AS total_sales
  FROM supply_chain_orders
  GROUP BY Category_Name, Product_Name
)
SELECT Category_Name, Product_Name, total_sales,
  RANK() OVER (PARTITION BY Category_Name ORDER BY total_sales DESC) AS rank_in_category
FROM product_sales
WHERE rank_in_category <= 3;
```

Business insight: Identifies the top 3 revenue drivers in each product category

26. Query 3: Month-over-month revenue growth rate

```
WITH monthly AS (
  SELECT DATE_FORMAT(order_date, "%Y-%m") AS month,
    SUM(Sales) AS revenue
  FROM supply_chain_orders
  GROUP BY month
)
SELECT month, revenue,
  LAG(revenue) OVER (ORDER BY month) AS prev_month,
  ROUND((revenue - LAG(revenue) OVER (ORDER BY month))
    / LAG(revenue) OVER (ORDER BY month) * 100, 1) AS mom_growth_pct
FROM monthly;
```

Business insight: Quantifies business acceleration/deceleration by month

27. Query 4: Customer lifetime value (CLV) with percentile ranking

```
WITH clv AS (
  SELECT Customer_Id,
    SUM(Sales) AS total_spent,
    COUNT(DISTINCT Order_Id) AS order_count
  FROM supply_chain_orders
  GROUP BY Customer_Id
)
SELECT Customer_Id, total_spent, order_count,
  NTILE(4) OVER (ORDER BY total_spent DESC) AS clv_quartile
FROM clv
ORDER BY total_spent DESC;
```

Business insight: Segments customers into top 25% vs bottom 25% by lifetime value

28. Query 5: Late delivery impact on revenue

```
WITH delivery_perf AS (
  SELECT
    CASE WHEN Days_for_shipping_real > Days_for_shipment_scheduled
         THEN "Late" ELSE "On Time" END AS delivery_flag,
    SUM(Sales) AS revenue,
    AVG(Order_Profit_Per_Order) AS avg_profit
  FROM supply_chain_orders
  GROUP BY delivery_flag
)
SELECT *, ROUND(revenue / SUM(revenue) OVER() * 100, 1) AS revenue_share_pct
FROM delivery_perf;
```

Business insight: Late deliveries still account for majority of revenue — operations bottleneck

29. Query 6: Top 5 regions by profit margin (CTE + RANK)

```
WITH region_kpi AS (
  SELECT Order_Region,
    SUM(Sales) AS total_sales,
    SUM(Order_Profit_Per_Order) AS total_profit,
    ROUND(SUM(Order_Profit_Per_Order)/SUM(Sales)*100, 2) AS profit_margin_pct
  FROM supply_chain_orders
  GROUP BY Order_Region
)
SELECT *, RANK() OVER (ORDER BY profit_margin_pct DESC) AS margin_rank
FROM region_kpi
ORDER BY margin_rank
LIMIT 5;
```

Business insight: Identifies most operationally efficient geographic markets

30. Query 7: Fraud detection: orders with suspicious profit ratios

```
SELECT Order_Id, Customer_Id, Sales,
  Order_Profit_Per_Order,
  ROUND(Order_Profit_Per_Order/Sales*100,1) AS margin_pct,
  Order_Status,
  AVG(Order_Profit_Per_Order) OVER (PARTITION BY Market) AS market_avg_profit
FROM supply_chain_orders
WHERE Order_Profit_Per_Order <
  (AVG(Order_Profit_Per_Order) OVER (PARTITION BY Market) - 2 *
   STDDEV(Order_Profit_Per_Order) OVER (PARTITION BY Market))
ORDER BY margin_pct ASC
LIMIT 20;
```

Business insight: Statistical outlier detection for fraud investigation — uses 2-sigma threshold

31. Query 8: Cohort: First order year retention (orders per customer year)

```
WITH first_order AS (
  SELECT Customer_Id, MIN(YEAR(order_date)) AS first_year
  FROM supply_chain_orders GROUP BY Customer_Id
),
cohort AS (
  SELECT f.first_year, YEAR(o.order_date) AS order_year,
    COUNT(DISTINCT o.Customer_Id) AS active_customers
  FROM supply_chain_orders o
  JOIN first_order f ON o.Customer_Id = f.Customer_Id
  GROUP BY f.first_year, order_year
)
SELECT first_year, order_year,
  order_year - first_year AS years_since_acquisition,
  active_customers
FROM cohort
ORDER BY first_year, order_year;
```

Business insight: Measures customer retention across acquisition cohorts — critical for CLV modeling

32. Query 9: ABC Analysis: Pareto of products by revenue

```
WITH product_rev AS (
  SELECT Product_Name, SUM(Sales) AS revenue
  FROM supply_chain_orders
  GROUP BY Product_Name
),
cumulative AS (
  SELECT Product_Name, revenue,
    SUM(revenue) OVER (ORDER BY revenue DESC) AS cum_revenue,
    SUM(revenue) OVER () AS total_revenue
  FROM product_rev
)
SELECT Product_Name, revenue,
  ROUND(cum_revenue/total_revenue*100, 1) AS cum_pct,
  CASE
    WHEN cum_revenue/total_revenue <= 0.8 THEN "A - High Value"
    WHEN cum_revenue/total_revenue <= 0.95 THEN "B - Mid Value"
    ELSE "C - Low Value"
  END AS abc_class
FROM cumulative
ORDER BY revenue DESC;
```

Business insight: Pareto ABC classification: ~20% of products drive 80% of revenue (A-class)

33. Query 10: Year-over-Year sales growth by market

```
WITH yearly AS (
  SELECT Market, YEAR(order_date) AS yr, SUM(Sales) AS revenue
  FROM supply_chain_orders
  GROUP BY Market, yr
)
SELECT Market, yr, revenue,
  LAG(revenue) OVER (PARTITION BY Market ORDER BY yr) AS prev_year,
  ROUND((revenue - LAG(revenue) OVER (PARTITION BY Market ORDER BY yr))
    / LAG(revenue) OVER (PARTITION BY Market ORDER BY yr) * 100, 1) AS yoy_growth
FROM yearly
ORDER BY Market, yr;
```

Business insight: Isolates which markets are growing vs stagnating year over year

34. Query 11: Dense rank customers by total spend (top customers)

```
SELECT Customer_Id,
  CONCAT(Customer_Fname, " ", Customer_Lname) AS customer_name,
  Customer_Segment,
```

```
  Market,
  ROUND(SUM(Sales), 2) AS lifetime_value,
  DENSE_RANK() OVER (ORDER BY SUM(Sales) DESC) AS value_rank
FROM supply_chain_orders
GROUP BY Customer_Id, customer_name, Customer_Segment, Market
ORDER BY value_rank
LIMIT 20;
```

Business insight: VIP customer identification for loyalty program targeting

35. Query 12: Percentile distribution of profit per order by segment

```
SELECT Customer_Segment,
  ROUND(MIN(Order_Profit_Per_Order), 2) AS min_profit,
  ROUND(PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY Order_Profit_Per_Order), 2) AS p25,
  ROUND(PERCENTILE_CONT(0.50) WITHIN GROUP (ORDER BY Order_Profit_Per_Order), 2) AS median,
  ROUND(PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY Order_Profit_Per_Order), 2) AS p75,
  ROUND(MAX(Order_Profit_Per_Order), 2) AS max_profit
FROM supply_chain_orders
GROUP BY Customer_Segment;
```

Business insight: Wide IQR for Corporate suggests inconsistent pricing/discounting

36. Query 13: Rolling 3-month average revenue

```
WITH monthly AS (
  SELECT DATE_FORMAT(order_date, "%Y-%m") AS month,
    SUM(Sales) AS revenue
  FROM supply_chain_orders
  GROUP BY month
)
SELECT month, revenue,
  ROUND(AVG(revenue) OVER (
    ORDER BY month ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
  ), 0) AS rolling_3mo_avg
FROM monthly
ORDER BY month;
```

Business insight: Smooths seasonality — management-ready trend chart baseline

37. Query 14: First and last order date per customer (activity span)

```
SELECT Customer_Id,
  MIN(order_date) AS first_order,
  MAX(order_date) AS last_order,
  DATEDIFF(MAX(order_date), MIN(order_date)) AS active_days,
  COUNT(DISTINCT Order_Id) AS total_orders
FROM supply_chain_orders
GROUP BY Customer_Id
HAVING total_orders > 1
ORDER BY active_days DESC
LIMIT 20;
```

Business insight: Active customer lifespan analysis — helps model churn risk

38. Query 15: Market contribution to overall profit (% share)

```
WITH market_p AS (
  SELECT Market, SUM(Order_Profit_Per_Order) AS profit
  FROM supply_chain_orders
  GROUP BY Market
)
SELECT Market, profit,
  ROUND(profit / SUM(profit) OVER() * 100, 1) AS profit_share_pct,
  RANK() OVER (ORDER BY profit DESC) AS rank_
FROM market_p;
```

Business insight: LATAM and Europe together contribute >60% of total global profit

SQL analysis coverage summary

The source document contains 38 query/code blocks identified in the SQL analytics section. They cover revenue and order counts, delivery performance, shipping mode comparisons, segment and market performance, product/category profitability, discount effects, time trends, customer value, cohort analysis, ABC/Pareto analysis, and fraud-risk screening.

PART 3 — FINAL RECOMMENDATIONS

- IMMEDIATE (0-30 days): Suspend Same Day shipping until SLA >95% achievable. Launch logistics partner audit.
- SHORT-TERM (30-90 days): Implement 15% hard cap on discounts. Deploy ML fraud scoring at checkout.
- MEDIUM-TERM (90-180 days): Fix data capture for Product Description and Order Zipcode. Build email CRM infrastructure.
- LONG-TERM (6-12 months): Expand into Africa with regional warehouse. Build predictive delivery model using this dataset.
- 30-Second Answer
- "I analyzed DataCo's global supply chain dataset — 180,000 records across 5 markets — using Python and SQL. I discovered that 57% of orders were delivered late and 19% of orders lost money due to excessive discounting. I built a full EDA with 20 visualizations and 40 SQL queries to surface these findings and proposed actionable recommendations to improve on-time delivery and restore profit margins."
- 2-Minute Answer
- "This project involved end-to-end analysis of a real-world supply chain dataset from DataCo Global, covering 65,752 unique orders across 5 global markets from 2015 to 2018. I acted as both a Data Analyst and a Business Analyst throughout."
- "My first step was a complete data quality audit. I discovered that 6 columns needed to be dropped — including customer passwords (security risk), a 100% null product description column, and an order zipcode field that was 86% missing. After cleaning, I engineered 12 new business features including delivery delay days, profit margin percentage, and a risk category flag."
- "In the EDA phase, I found three critical business problems: First, a 57.3% late delivery rate — nearly three times the industry benchmark. Second, 18.7% of orders produced negative profit, directly caused by unconstrained discount policies where any discount above 15% consistently destroyed margin. Third, 2.25% of orders were flagged as suspected fraud, representing roughly $830K in annual revenue leakage."
- "I then designed a SQL star schema and wrote 40+ queries from basic aggregations to advanced window functions like running totals, LAG for MoM growth, and NTILE for customer CLV percentile ranking. My three key recommendations were: immediately audit the Same Day shipping SLA, implement a 15% discount cap, and deploy ML-based fraud detection at the payment stage."
- 5-Minute Detailed Presentation
- Use the following structure:
- Context (30s): DataCo Global, 5 markets, 4 years, 180K records, $36.8M revenue
- Data Audit (60s): 6 dropped columns, 3 critical quality issues, justification for each decision

Action plan synthesized from the report

Priority	Action	Why it matters / suggested measure
1 — Delivery reliability	Audit carrier and shipping-mode performance; investigate the reported high late-delivery rate, especially premium modes.	Track late-delivery rate by mode, market, region and carrier; set an improvement target and weekly review.
2 — Profitability controls	Review negative-profit SKUs and orders; introduce discount guardrails and floor-price checks.	Track negative-profit order rate, margin by category, and profit by discount band.
3 — Fraud-risk review	Use the suspected-fraud flag and	Track suspected-fraud rate and

	anomalous profit/order patterns to prioritize manual review; assess a fraud-scoring model as a future extension.	confirmed loss after validation; do not treat a dataset flag as proof of fraud.
4 — Seasonal planning	Prepare inventory, staffing and carrier capacity ahead of recurring Q4 peaks identified in the report.	Compare forecast versus actual monthly sales, stockouts and on-time delivery during peak periods.
5 — Market and product concentration	Protect key revenue-driving products while evaluating regional performance and expansion opportunities using profitability as well as sales.	Monitor market revenue, profit margin, SKU concentration and stock availability.
6 — Order-status bottlenecks	Investigate pending-payment, on-hold and cancellation statuses; consider escalation rules for aged orders.	Track order-status aging, cancellation rate and time to completion.
7 — Data governance	Improve capture and validation of product descriptions, customer contact fields and postal/geographic data.	Track completeness and validity rates for critical fields; avoid retaining unnecessary sensitive fields.
8 — Reporting and ownership	Build a consistent KPI dashboard with definitions, owners, refresh cadence and targets.	Include revenue, profit, margin, late delivery, average shipping days, fraud flag rate, negative-profit rate and customer counts.

Interpretation and validation notes

- Some figures in the report vary across sections (for example, late-delivery share is shown as 57.3% in the KPI summary and 54.8% in one delivery-status chart description). Recalculate metrics from the dataset and document the denominator before presenting them.
- The reported $830K fraud exposure and claims about discount thresholds should be treated as estimates/hypotheses until the calculation method and underlying records are validated.
- The source report covers January 2015 to February 2018; avoid describing the results as current operational performance.
- SQL snippets are preserved from the supplied document. Confirm that the table/column names and SQL dialect match the database used to execute them.